RANGKUMAN BAB 7

IoT Communications Technologies

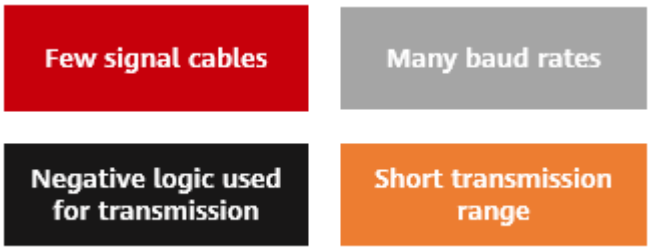
1. Wired Communications Technologies

Ethernet

- Ethernet is a computer local area network (LAN) technology. Technical standard: Institute of Electrical and Electronics Engineers (IEEE) 802.3, which defines cable connections at the physical layer, electronic signals, and protocols at the media access control (MAC) layer.
- A base station has three types of ports: one ETH port, two FE/GE ports, and two SFP ports. SFP ports are gigabit optical ports. ETH is a standard Ethernet network, which transmits data at the rate of 10 Mbit/s. Fast Ethernet (FE) supports a transmission rate of 100 Mbit/s. Gigabit Ethernet (GE) supports a transmission rate of 1,000 Mbit/s. FE/GE are auto-adaptive ports that support 100 Mbit/s and 1,000 Mbit/s based on auto negotiation with the peer switch.
- Ethernet uses carrier sense multiple access with collision detection (CSMA/CD). Carrier sense: The carrier is checked before sending. Multiple access: Data sent by a station is received by other stations. Collision detection: The station detects whether there is a collision when sending data.
- CSMA/CD is a data transmission method used in Ethernet.
- Conflict detection: When a station sends a frame, it listens to the media to detect whether a collision occurs (whether other stations are sending frames.)
- CSMA/CD specifications are standardized by IEEE 802.3 and ISO 8802-3.
- IEEE 802.3 defines the Ethernet physical (PHY) layer, which is a transceiver for transmitting and receiving data at the physical layer. The physical layer defines electrical and optical signals, link status, clock reference, data coding, and circuits required for data transmission and reception, and provides standard interfaces for devices at the data link layer.
- IEEE 802.3 defines the Ethernet MAC, which is a media access controller and implements the data link layer. The data link layer provides functions such as addressing mechanism, data frame construction, data error check, transmission control, and standard data interfaces for the network layer.

RS-232

- RS-232 (or EIA RS-232) is a standard interface between data terminal equipment (DTE) and data communications equipment (DCE) for serial binary data exchange. It was jointly developed in 1970 by the Electronics Industries Association (EIA), Bell System, modem manufacturers, and computer manufacturers.
- Technical features:



RS-485

- Originally: Simple process values were output based on analog signals. Later: RS-232 was used as an instrument interface for point-to-point communications, but cannot implement connections at scale. Solution: RS-485 was developed.
- RS-485 is a serial communications protocol without the disadvantages of RS-232. RS-485 uses a differential, half-duplex transmission mode and supports point-tomultipoint communications. An RS-485 bus supports two-wire and four-wire connections. Four-wire connections implement only point-to-point communications thus are seldom used. Two-wire connections support both point-to-point (singleslave) and point-to-multipoint (multi-slave) communications.
- In the early 1980s, MCU technologies were maturing, and smart meters dominated the global meter market. This is driven by enterprise digitalization. Enterprises want meters with network communication interfaces.

RS-232 Vs RS-485

Item	RS-232	RS-485
Transmission range	Less than 20 m	Theoretical:1,200 m; actual: 300–500 m
Transmission mode	Unbalanced, single-end	Balanced, differential
Number of transceivers	Point-to-point	Up to 128 transceivers on the bus
Transmission rate	38.4 Kbit/s	10 Mbit/s

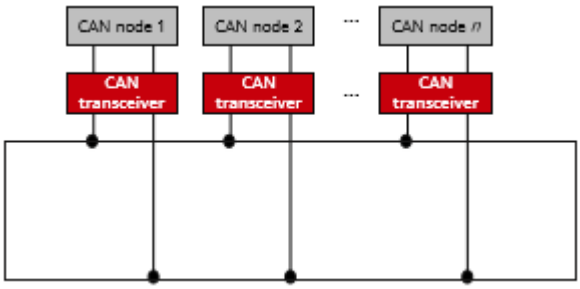
- If serial communications are compared to traffic and UART is compared to a station, a frame of data is like a car. Cars on the road must follow traffic rules. The speed limit can be 30 km/h or 40 km/h in a city, and 120 km/h on highways. What road a car runs on and how fast a car runs depend on the protocol. Common serial interface protocols include RS-232, RS-422, and RS-485.

CAN

- Controller Area Network (CAN) is a serial communications protocol recognized by the International Organization for Standardization (ISO). It was developed by Bosch (a

German company) in 1983 for data communications between monitoring and execution modules of an automobile internal control system. It is now one of the most widely used fieldbuses in the world.

- A CAN bus uses serial data transmission. When a node on the CAN bus broadcasts data packets, all other nodes (including non-targets) receive the data.
- CAN bus topology:



- Main features of the CAN bus:
    1) Long data transmission range (up to 10 km)
    2) High data transmission rate (up to 1 Mbit/s)
    3) Excellent arbitration mechanism
    4) It uses filters to implement multi-address data frame delivery.
    5) It uses remote frames for remote data requests.
    6) It can detect and process errors.
    7) It can automatically resend data.
    8) A faulty node can be automatically disconnected from the bus without affecting other nodes.

USB

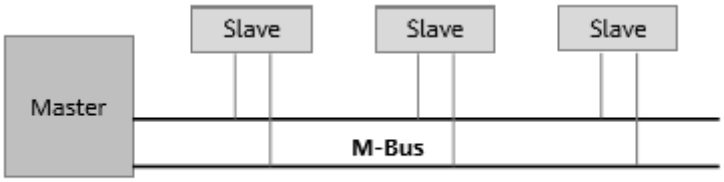
- Universal Serial Bus (USB) is a standard for connecting computer systems and external devices. It is also a technical specification for input/output interfaces widely used on PCs, mobile devices, cameras, set top boxes (STBs), and game consoles.
- USB rates of each version:

Official Version	Original Name	Transmission Rate
LowSpeed	USB 1.0	1.5 Mbit/s
FullSpeed	USB 1.1	12 Mbit/s
HiSpeed	USB 2.0	480 Mbit/s
Gen 1	USB 3.0 Gen1	5 Gbit/s
Gen 2	USB 3.0 Gen2	10 Gbit/s
Gen 2×2	N/A	20 Gbit/s
N/A	USB 4.0	40 Gbit/s

- USB is designed for high speed, scalability, and ease of use.
- USB is much faster than standard buses used by traditional computers,

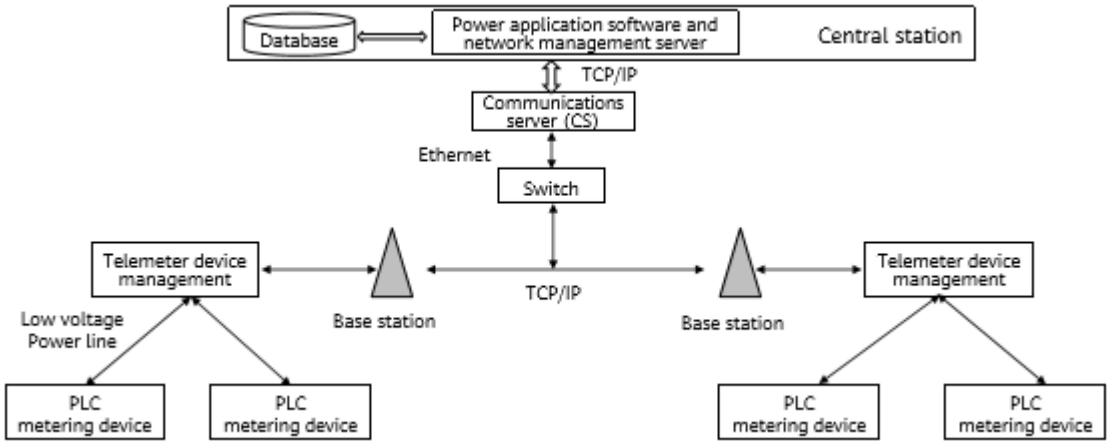
M-Bus

- Meter-Bus (M-Bus) is a standard for data buses of measuring instruments and counters to transmit information. It is widely used for collecting energy consumption data of commercial and industrial buildings.
- Its transmission limit is 1,000 m and can supply power to field devices without auxiliary power cables. A bus allows up to 5 A voltage, and a node allows up to 0.65 mA power.
- M-Bus topology:



PLC

- Power Line Communication (PLC) is a communication technology that transmits data and media signals over power lines. PLC loads high frequency signals containing information onto the current. An adapter receives the information over the cable, separates the highfrequency signals from the current, and sends the signals to a computer or telephone.



➢ The parameters of the three PLC technologies are as follows:

1) G3-PLC: subcarrier spacing: 1.5625 kHz; subcarrier width: 3.125 kHz; transmission bandwidth: 54.7 kHz. A total of 36 subcarriers can be deployed.
2) PRIME: subcarrier spacing: 0.488 kHz; subcarrier width: 0.976 kHz; transmission bandwidth: 47 kHz. A total of 96 subcarriers can be deployed.
3) PLC-IoT: subcarrier spacing: 4.6875 kHz; subcarrier width: 9.375 kHz; transmission bandwidth: 2,000–12,000 kHz. A total of 128 subcarriers can be deployed.

Comparison of Wired Communications Technologies

Mode	Feature	Application Scenario
Ethernet	Comprehensive protocols, universal, cost-effective	Intelligent devices, video surveillance
RS-232	Point-to-point communications, cost-effective, short transmission range	A few instruments, industrial control
RS-485	Bus topology, cost-effective, strong anti-interference capability	Industrial instruments, metering
CAN	High real-time performance, long transmission range, vulnerable to electromagnetic interference, and cost-effective	Industrial automation, ships, medical equipment, industrial equipment
USB	Point-to-point communications, universal, fast transmission	Smart home, office, mobile devices
M-Bus	Designed for metering, common twisted-pair cables, strong anti-interference capability	Industrial energy consumption data collection
PLC	For power line communications, wide coverage, easy installation	Power grid transmission, electricity meters

2. Wireless Communications Technologies

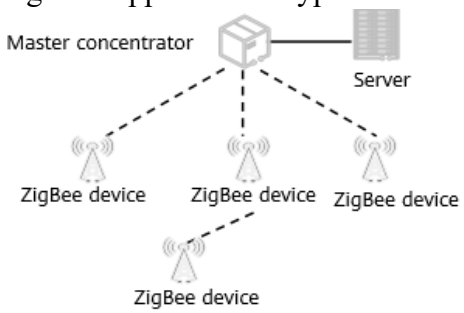
➢ Short-Range Wireless Communicatins Technologies

Bluetooth

- Bluetooth is a large-capacity wireless communications technology standard for short distances. The latest Bluetooth 5.0 has a transmission limit of 3 Mbit/s and 300 meters. Two types of Bluetooth: Bluetooth Basic Rate/Enhanced Data Rate (BR/EDR) and Bluetooth Low Energy (BLE). BR/EDR supports only point-to-point (one-to-one) communications. BLE supports point-to-point, broadcast (one-tomany), and mesh (many-to-many) communications. BLE is mainly used in IoT for smart home products to deliver better performance and consume less power.
- Advantages: high rates, high security, and low power
- Disadvantages: EDR has few network nodes thus is not suitable for multi-point deployment.
    - ➢ Bluetooth is a wireless communications technology standard that implements short-range data exchange between fixed devices, mobile devices, and personal area networks (PDNs). It uses ISM 2.4–2.485 GHz ultra high frequency (UHF) bands. Bluetooth was developed by the telecom giant Ericsson in 1994 as an alternative to RS-232 data lines. It can connect multiple devices, without data synchronization issues.
    - ➢ Today, Bluetooth is managed by the Bluetooth Special Interest Group (SIG). Bluetooth SIG has more than 25,000 members around the world in different industries such as telecommunications, computers, networks, and consumer electronics.
    - ➢ Bluetooth 5.0 supports faster and longer transmission with low power consumption. The transmission rate is twice that of Bluetooth 4.2 (up to 2 Mbit/s). The transmission range is four times that of Bluetooth 4.2 (theoretically up to 300 m). The data packet capacity is eight times that of Bluetooth 4.2.
    - ➢ It supports indoor positioning and navigation and can achieve a positioning precision of less than 1 m by working with Wi-Fi.
    - ➢ It optimizes IoT foundation layers to enable lower power consumption and higher performance for smart homes.

ZigBee

- ZigBee is a short-range, low-rate wireless communications technology with low power. It is for periodic data, intermittent data, and low response time data transmission. Features: short transmission range, low complexity, self-organization, low power, and low data rate. It is widely used in industrial and smart home fields.
- ZigBee supports three types of network topologies: star, tree (cluster), and mesh.

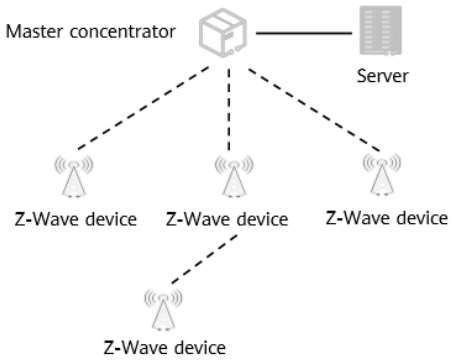


ZigBee	
Low power	Two batteries for 6–24 months
Low cost	No patent fee. Cost of each module: USD2
Low rate	20–250 Kbit/s
Short range	10–100 m
Low latency	15–30 ms
Large capacity	Up to 254 nodes
Strong security	Three-layer security
License-free	915 MHz, 868 MHz (Europe) 2.4 GHz (global)
Easy networking	Mesh networking, self-networking
Incompatibility	Incompatible between different chips
Difficult to maintain	Too flexible to maintain

- ZigBee is a low-power local area network (LAN) protocol based on IEEE 802.15.4 and also a short-range, low-power wireless communications technology. The name is derived from the waggle dance of honey bees after their return to the beehive. Bees exchange the source of pollen location to by zig-zagging and other moves. In other words, the waggle dance helps bees to build their communication network. Features: short transmission range, low complexity, selforganization, low power, and low data rate. ZigBee can be integrated with various devices for automatic control and remote control.

Z-Wave

- Z-Wave is an emerging short-range wireless communications technology that uses radio frequency (RF) to transmit data.
- Advantages: simple structure, low rates, low power, low cost, and high reliability
- Disadvantages: Closed standard. The chip can only be obtained from Sigma Designs.



  - Z-Wave is an emerging short-range wireless communications technology that uses radio frequency (RF) to transmit data. It features low power

consumption, low costs, and high reliability. It operates at the 868.42 MHz (in Europe) to 908.42 MHz (in the US) frequency bands and uses binary or Gaussian frequencyshift keying (FSK) modulation. Z-Wave can transmit data at 9.6 kbit/s. Its transmission range is 30 m indoors and 100 m outdoors. It is perfect for broadband and narrowband scenarios.

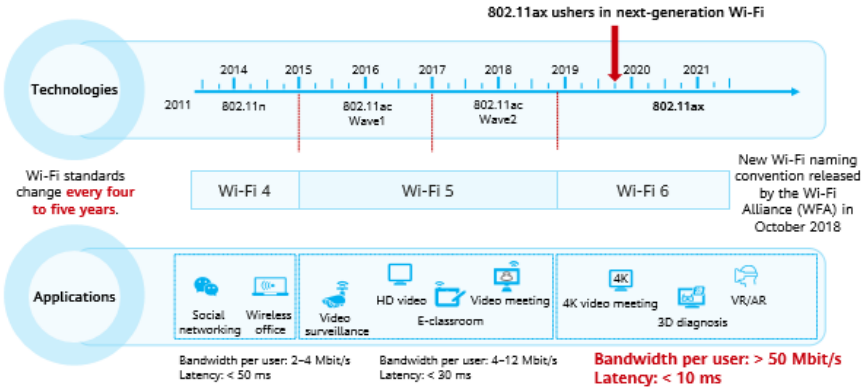
- Z-Wave is used for residential and lighting control and status reading scenarios, such as metering, lighting and home appliance control, access control, anti-theft, and fire detection.
- Z-Wave was designed for smart home wireless control. A small data format is used for transmission, so 40 kbit/s is enough. In the early stage, data was even transmitted at 9.6 kbit/s. Compared with other similar wireless communications technologies, it has a relatively low transmission frequency, long transmission range, and price advantages.

Wi-Fi

- Wi-Fi connects electronic devices to a wireless local area network (WLAN) using the 2.4 GHz ultra high frequency (UHF) or 5 GHz super high frequency (SHF) industrial, scientific, medical (ISM) radio frequency band. The latest Wi-Fi 6 supports a transmission rate of 9.6 Gbit/s and a latency of 20 ms.
- Advantages: wide coverage, fast data transmission
- Disadvantages: low transmission security, low stability, high power consumption, and weak networking
- Access controller (AC)
- Access point (AP)
- Station (STA) client
    - 802.11ax supports all ISM frequency bands from 1 GHz to 6 GHz, including the currently used 2.4 GHz and 5 GHz frequency bands. It is compatible with 802.11a, 802.11b, 802.11g, 802.11n, and 802.11ac. It aims to support indoor and outdoor communication scenarios and improve spectral efficiency. Compared with 802.11ac, 802.11ax increases the actual throughput by four times, improves the nominal transmission rate by 37%, and reduces the latency by 75% in dense user environments.
    - Main advanced functions:
        1) It is compatible with 802.11a, 802.11b, 802.11g, 802.11n, and 802.11ac.
        2) Orthogonal Frequency Division Multiple Access (OFDMA): enables channel sharing for uplink and downlink data transmission in strict environments to improve network efficiency and reduce latency.
        3) Multi-user multiple-input multiple-output (MU-MIMO): allows more downlink data to be transmitted at a time, so that the access point can transmit data to more devices.
        4) 160 MHz channel: The bandwidth is increased to provide higher performance with low latency.
        5) 1024-quadrature amplitude modulation (1024-QAM): More data is encoded in the same amount of spectrum to improve the throughput of Wi-Fi devices.

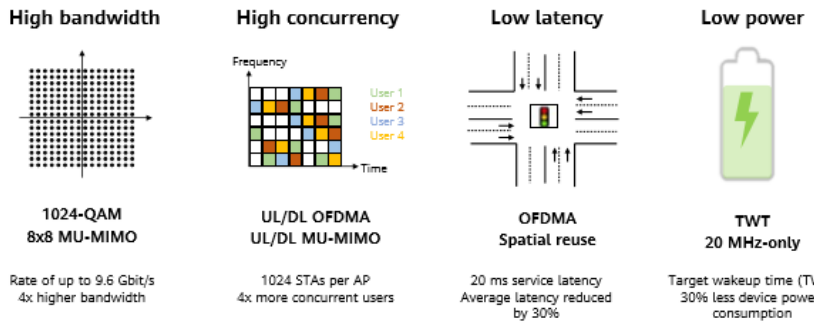
6) Target wake-up time (TWT): significantly prolongs the battery life of Wi-Fi devices, such as IoT devices.
7) Transmit beamforming: supports a higher data rate within a given range, thereby providing a larger network capacity.
8) Four times the OFDM symbol duration.
9) Adaptive clear channel assessment (Adaptive CCA).
10) The security standard is upgraded to Wi-Fi Protected Access (WPA) 3.

Technology and Application Development Promotes Wi-Fi 6



- Wi-Fi 5 cannot meet the low service latency and high bandwidth requirements of 4K/8K video meeting scenarios.
- Huawei Wi-Fi 6 uses SmartRadio intelligent application acceleration to reduce latency to as low as 10 ms, thereby meeting such requirements.

Features of Wi-Fi 6



- The bandwidth is increased by four times compared with the theoretical rate. Currently, the theoretical rate (in Wave2) of all Wi-Fi 5 products is 2.5 Gbit/s, and the theoretical rate of Wi-Fi 6 products is 9.6 Gbit/s.
- The number of concurrent users supported is increased by four times. In the actual test, Wi-Fi 5 supports 100 concurrent users and Wi-Fi 6 supports 400 concurrent users when the per capita bandwidth is 2 Mbit/s.
- The average latency of Wi-Fi 6 is about 20 ms, while the average latency of Wi-Fi 5 is about 30 ms. After Huawei SmartRadio intelligent application acceleration is used, the service latency can be reduced to 10 ms. • Wi-Fi 5 does not support TWT.

IEEE 802.11 Standards and Wi-Fi Generations

Wi-Fi	802.11 Standard	Released In	Frequency Band	Theoretical Rate
-	802.11	1997	2.4 GHz	2 Mbit/s
-	802.11 b	1999	2.4 GHz	11 Mbit/s
-	802.11 a	1999	5 GHz	54 Mbit/s
	802.11 g	2003	2.4 GHz	54 Mbit/s
Wi-Fi 4	802.11 n	2009	2.4 GHz or 5 GHz	600 Mbit/s
Wi-Fi 5	802.11 ac Wave1	2013	5 GHz	3.47 Gbit/s
	802.11 ac Wave2	2015	5 GHz	6.9 Gbit/s
Wi-Fi 6	802.11 ax	2018/2019	2.4 GHz or 5 GHz	9.6 Gbit/s

• The first version of the IEEE 802.11 standard was released in 1997 to define the MAC and PHY layers.

• Since then, supplemental standards based on 802.11 have been defined. The most well-known ones influenced Wi-Fi evolution over generations: 802.11b, 802.11a, 802.11g, 802.11n, and 802.11ac.

Comparison of Short-Range Wireless Communications Technologies

	Bluetooth	Wi-Fi	ZigBee	Z-Wave
Frequency band	2.4 GHz	2.4 GHz, 5 GHz	868 MHz, 915 MHz, 2.4 GHz	868.42 MHz (Europe), 908.42 MHz (US)
Transmission rate	1–3 Mbit/s (24 Mbit/s over 802.11 links)	802.11b: 11 Mbit/s 802.11g: 54 Mbit/s 802.11n: 600 Mbit/s 802.11ac: 1 Gbit/s 802.11ax: 9.6 Gbit/s	868 MHz: 20 Kbit/s 915 MHz: 40 Kbit/s 2.4 GHz: 250 Kbit/s	9.6 Kbit/s or 40 Kbit/s
Transmission range	1–300 m	50–100 m	2.4 GHz band: 10–100 m	30 m (indoor) to 100 m (outdoor)
Scenarios	Data exchanged by nearby nodes (mouse, wireless headset, mobile device, and computer)	WLAN, high-speed Internet access at homes and other indoor places	Home automation, building automation, remote control	Smart home, monitoring and control

2. Wireless Communications Technologies

Cellular Mobile Communications Technologies

2G

1. GSM Standard (2G):
    - Developed in 1992 by the European Committee for Standardization.
    - Aims to unify digital communication technologies into a network standard.
    - Ensures communication quality and facilitates new services.
    - Data rate: 9.6 Kbit/s.
2. GPRS (General Packet Radio Service):
    - Mobile data service for GSM mobile phone users.
    - Extension of GSM, providing data rates of 56–114 Kbit/s.
3. 2G Mobile Phone Communication Technologies:
    - GSM:
        - Developed based on Time Division Multiple Access (TDMA).
        - Originated in Europe and globally adopted.
    - iDEN (Integrated Digital Enhanced Network):

        - Developed based on TDMA.
        - Used by Nextel, a US telecommunications provider.
    - IS-136 (or D-AMPS):
        - Developed based on TDMA.
        - Simplest TDMA system in the US, used in the Americas.
    - IS-95 (or CDMA One):
        - Developed based on Code Division Multiple Access (CDMA).
        - Simplest CDMA system in the US, used in the Americas and some Asian countries.
    - PDC (Personal Digital Cellular):
        - Developed based on TDMA.
        - Exclusively used in Japan.

3G

1. Definition of 3G:
    - Third-generation mobile communications technology.
    - Enables high-rate data transmission, supporting voice and data simultaneously at hundreds of Kbit/s.
    - Integrates wireless and multimedia communications, including Internet access.
2. 3G Standards:
    - Three main standards: CDMA2000, WCDMA, and TD-SCDMA.
    - HSPA+, the latest WCDMA technology, offers a downlink rate of up to 42 Mbit/s.
3. International Telecommunication Union (ITU) Standard Name:
    - Known as International Mobile Telecom System-2000 (IMT-2000) in the ITU standard.
    - Originally proposed as the Future Public Land Mobile Telecommunication System (FPLMTS) in 1985.
    - Renamed to IMT-2000 in 1996, indicating operation at the 2,000 MHz frequency band and a maximum rate of 2,000 Kbit/s.
    - Commercially used around 2000.
4. Background and Transition from 2G:
    - 3G arose due to the success of 2G, leading to increased user growth and capacity challenges.
    - Standardization efforts for 3G began in 1997.

- Initially named Universal Mobile Telecommunication System (UMTS) by the European Telecommunications Standards Institute (ETSI).

4G

1. Definition of 4G:
    - Fourth-generation mobile communications technology.
2. Modes of 4G:
    - Includes two modes: LTE time division duplexing (TDD) and LTE frequency division duplexing (FDD).
3. Capabilities of 4G:
    - Integrates 3G and WLAN.
    - Transmits data, high-quality audio, videos, and images at high rates.
    - Download rate exceeds 100 Mbit/s, 25 times faster than ADSL (4 Mbit/s).
    - Meets diverse user requirements for wireless services.
    - Deployment in areas without DSL and cable television modem coverage.
4. ITU Standard Name:
    - Officially named International Mobile Telecommunications-Advanced (IMT-Advanced) by the ITU.
5. Advanced Features of 4G:
    - Carrier Aggregation (CA) enables a maximum downlink rate of 330 Mbit/s or higher.
    - Voice over LTE (VoLTE) allows 4G users to avoid falling back to 2G or 3G networks during voice calls.
    - VoLTE improves voice quality by 40% compared to 2/3G, with a 50% decrease in delay compared to 3G networks.
6. ITU Requirements and Evolution:
    - ITU requires a rate of 1 Gbit/s for 4G.
    - Initial rates like 100 Mbit/s (TD-LTE) or 150 Mbit/s (FDD-LTE) do not meet the requirement.
    - LTE-Advanced achieved 600 Mbit/s, and LTE-Advanced Pro achieved 1 Gbit/s through continuous evolution.

LTE Cat

Level	Downlink Rate (Mbit/s)	Downlink MIMO	Uplink Rate (Mbit/s)
1	10	1	5
2	50	2	25
3	100	2	50
4	150	2	50
5	300	4	75
6	300	2 or 4	50
7	300	2 or 4	150
8	1,200	8	600
9	450	2 or 4	50
10	450	2 or 4	100

1. LTE Cat and UE:
    - LTE Cat (LTE User Equipment Category) indicates the access capability level or transmission rate supported by a UE (User Equipment) in a 4G LTE network.
2. User Equipment (UE):
    - UE refers to a device used by a user to access or use a 4G LTE network.
    - Examples include 4G mobile phones, smartwatches, or tablets.
3. LTE Cat Levels:
    - LTE Cat 4, Cat 6, and Cat 9 signify access capability levels 4, 6, and 9 of UE LTE networks, respectively.
4. Maximum Transmission Rate:
    - LTE categories specify the theoretical upper limit of UE transmission rates.
    - The actual rates depend on network conditions, including optimization and the number of UEs.
    - Actual rates can approach the upper limit but may not reach it.
5. Dependency on Network Conditions:
    - Actual transmission rates are influenced by factors such as network optimization and the number of UEs in the network.
    - Theoretical limits are indicative, and real-world rates can only approach, not necessarily reach, these upper limits.

LTE Cat 1

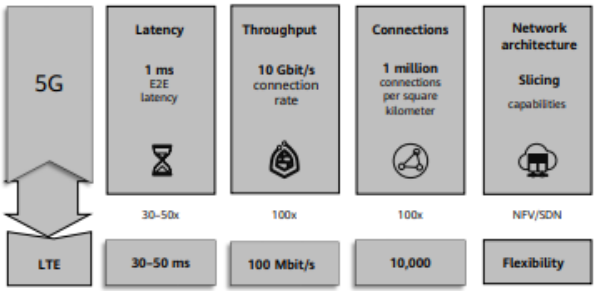
1. LTE Cat 1 Overview:
    - Supports up to 10 Mbit/s downlink for IoT devices.
    - Enables cost-effective and low-power connectivity to LTE networks.
2. Compatibility and Deployment:
    - LTE operators globally deploy networks based on 3GPP Release 8 or later.
    - LTE Cat 1 terminals can access networks without requiring a costly upgrade.
3. Cost-Effectiveness:

- Despite higher-rate alternatives like LTE Cat 4, LTE Cat 1 is preferred for IoT due to its affordability without compromising essential performance.

5G

1. 5G Overview:
    - Fifth-generation mobile communications technology.
    - Theoretical transmission rate of 10 Gbit/s, 100 times faster than 4G.
    - Enables downloading a 1 GB movie in eight seconds.
2. ITU-R Scenarios and Capabilities:
    - ITU-R defined three major 5G application scenarios: eMBB, mMTC, and uRLLC.
    - Eight capability specifications include throughput, latency, connection density, and spectral efficiency.
3. 5G Commercialization:
    - MIIT granted 5G licenses for commercial use to major Chinese telecom providers in June 2019.
    - 5G services officially launched on October 31, 2019.
4. IMT-2020 Standard:
    - 5G officially named IMT-2020 by ITU.
5. Key Capability Indicators:
    - Peak rate of 20 Gbit/s, 100 Mbit/s data rate for users, three times higher spectral efficiency than IMT-A.
    - 500 km/h mobility, 1 ms latency, 1 million connections per square kilometer (connection density).
    - 100 times the energy efficiency of IMT-A, and 10 Mbit/s per square meter of traffic density.

5G: Performance Indicators



4G cannot accommodate future applications and requirements, unable to meet their requirements for latency, throughput, and connections.

5G: Three Major Innovations

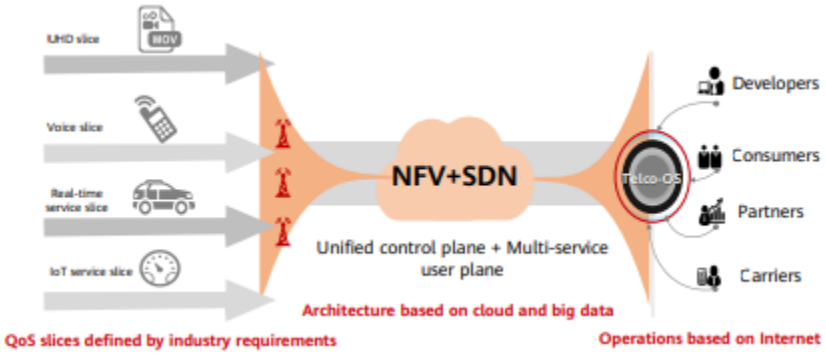


New Architecture: High-band and low-band aggregation for better experience

New Radio: One physical network for hundreds of industries

Full Spetrum: Adaptability to various services for higher spectral efficiency

5G Architecture: One Network for Hundreds of Industries



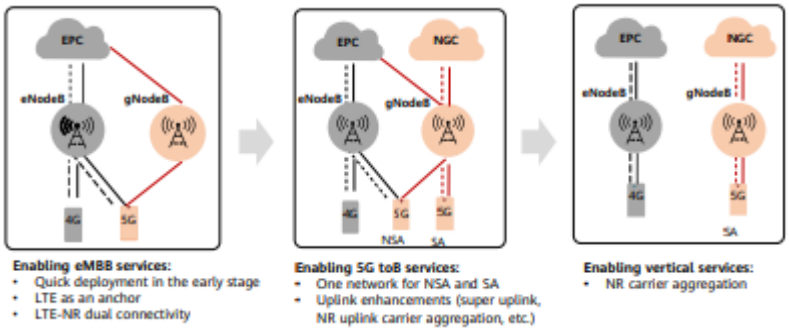
1. Network Function Virtualization (NFV):
    - Utilizes universal hardware and virtualization to reduce network device costs.
    - Decouples software and hardware, allowing flexible resource sharing.
    - Enables quick development and deployment of new services.
    - Supports automatic deployment, auto scaling, fault isolation, and self-healing based on service requirements.
2. Software-Defined Networking (SDN):
    - Innovative architecture proposed by Stanford University's Clean Slate Program.
    - Core technology, OpenFlow, separates control and data planes for flexible network traffic control.
    - Enhances network intelligence, providing a platform for core network and application innovation.
3. Network Slicing:
    - Builds multiple virtual end-to-end networks on universal hardware.
    - Each network slice caters to different service requirements, slicing physical resources (access, connection, computing, and storage).

- Key features include customization, end-to-end coverage, and isolation.

4. Key Features of Network Slices:
    - End-to-end network covering RAN, transport network, and core network.
    - Requires a cross-domain slice management system.
    - Demands isolation of resources, security, and OAM, achievable through different technologies.
    - Customizable to provide specific network functions and features.

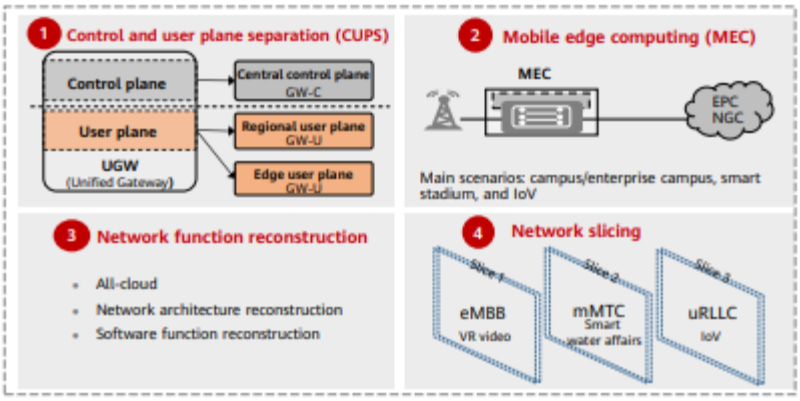
5G Networking

2019 NSA Option3X - 2020–2023 Long-term coexistence of NSA and SA - 2023+ SA as the target architecture



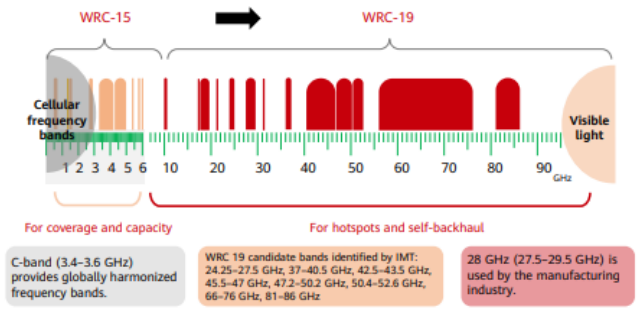
1. Multiple Stream Aggregation (MSA):
    - Terminal can use multiple base stations with the same or different RATs for data transmission.
2. Non-Standalone (NSA):
    - 4G core network + 4G/5G base station.
    - Independent user plane, no independent control plane over NR.
3. Standalone (SA):
    - 5G core network + new 5G base station.
    - Independent control and user planes.
4. eNB (Evolved NodeB):
    - Evolved 4G base station.
5. gNB (Next-Generation NodeB):
    - Next-generation 5G base station.

NGC: Four Service-oriented Features



1. Next Generation Core (NGC):
    - Represents the 5G core network.
2. Mobile Edge Computing (MEC):
    - Offers cloud computing capabilities on the Radio Access Network (RAN) close to mobile users.
    - Provides low latency, high bandwidth, and real-time information about wireless networks.
    - Enables application developers and content providers to deliver differentiated services based on user locations and base station loads.

5G: Spectrum Aggregation



1. Sub-3 GHz:
    - Frequencies below 3,000 MHz.
    - Primarily used for coverage.
2. C-band:
    - Frequencies between 3,000–6,000 MHz.
    - Mainly used for capacity expansion.
3. mmWave:
    - Primarily used for hotspots and self-backhaul.
4. Full Spectrum:

- Low-frequency bands below 6 GHz for connectivity, coverage, mobility, and basic capacity.
- High-frequency bands (>6 GHz) for higher data rates.

1. 5G Frequency Allocation in China:
    - MIIT licensed low and mid-frequency bands for commercial use by major carriers.
    - China Mobile, China Telecom, and China Unicom were allocated specific frequency bands.
2. 5G Use Cases:
    - Enhanced Mobile Broadband (eMBB) for faster transmission, supporting VR, AR, and MR.
    - Massive Machine Type Communications (mMTC) for IoT connections, including NB-IoT.
3. Ultra-Reliable Low Latency Communications (uRLLC):
    - 5G provides ultra-high reliability and low-latency services for applications like assisted driving and remote surgery.
4. Comparison of Cellular Mobile Communications Technologies:
    - Contrasts 2G, 3G, 4G, and 5G in terms of licensed frequency bands, transmission rates, and supported scenarios.
5. LPWA Technologies Overview:
    - LPWA technologies, including Sigfox, LoRa, NB-IoT, and eMTC, are designed for IoT applications with diverse requirements.
    - LPWA technologies offer a range of data rates, coverage distances, and use cases to accommodate the needs of different IoT applications.
6. Sigfox:
    - Sigfox utilizes Ultra Narrowband (UNB) for stable transmission with low power.
    - It operates in unlicensed ISM radio bands at frequencies like 868 MHz in Europe and 915 MHz in the United States.
    - Sigfox's advantages include free bands, low power consumption, and a lean network architecture.
7. LoRa:
    - LoRa (Long Range) employs chirp spread spectrum (CSS) modulation for long-range data transmission.
    - It uses unlicensed spectrum and is maintained by the LoRa Alliance.
    - LoRa is suitable for various applications, including automatic metering, smart home appliances, industrial monitoring, and more.
8. eMTC (enhanced Machine Type Communication):
    - eMTC is a wireless IoT solution proposed by Ericsson, developed based on LTE.
    - It offers higher data rates (up to 1 Mbit/s) and power consumption compared to NB-IoT but has smaller coverage and capacity.
    - eMTC supports voice transmission and is designed for low-rate, wide-coverage, low-power, and massive connection scenarios.
9. NB-IoT (Narrowband IoT):

- NB-IoT is built on cellular networks, consumes only 180 kHz bandwidth, and can be embedded into 5G.
- It features enhanced coverage, low cost, low power, and massive connection capabilities.
- NB-IoT has become a mainstream technology in cellular IoT, supporting diverse applications like remote metering, asset tracking, smart parking, and agriculture.

10. Key Features of NB-IoT:
    - Enhanced coverage, low cost, low power, and massive connections.
    - Supports LTE-based smooth evolution, flexible deployment, and low-cost communication chips.
11. Power Efficiency in NB-IoT:
    - Describes power-saving features like DRX (discontinuous reception), eDRX, and PSM (Power Saving Mode).
12. Enhanced Coverage in NB-IoT:
    - Explains increased power spectral density and time-domain retransmission for improved coverage.
13. Massive Connections in NB-IoT:
    - Discusses reduced air interface signaling overhead and resources for efficient handling of numerous IoT devices.
14. Comparison of LPWA Technologies:
    - LPWA technologies cater to different IoT scenarios based on data rates, coverage, and power consumption.
    - Cat 1 (Category 1) LTE is highlighted as a technology offering a balance between communication functions, cost-effectiveness, and integration with existing LTE networks.
    - Compares Sigfox, LoRa, NB-IoT, and eMTC in terms of frequency band, transmission rate, range, and use cases.
15. Comparison of Wireless Communications Technologies:
    - Compares data rates, coverage, and use cases of short-range (Z-Wave/ZigBee), high-rate short-range (Wi-Fi/Bluetooth), high-rate long-range (4G/5G), low-rate long-range (2G), and LPWA technologies.
16. IoT Network Capacity Requirements:
    - Estimates that 10% of IoT nodes require high bandwidth (4G/5G), 30% require medium bandwidth (Cat 1, eMTC), and 60% require low bandwidth (NB-IoT, LoRa).
17. Advantages of Cat 1:
    - Better communication functions than NB-IoT, lower costs than eMTC and Cat 4, and seamless access to existing LTE networks.

Quiz

1) (True or false) All NB-IoT networks are deployed on licensed sub-GHz bands.
   False
2) (Single-answer question1) Which of the following is a wired communications technology?
   A. 5G
   B. NB-IoT
   C. PLC
   D. ZigBee